

# A Mortgage-Specific Safety-Guardrail Benchmark, Grounded in Real HMDA Data

Reza Rahimi

JazzX AI, Los Altos, CA, USA

reza.rahimi@jazzx.ai

Working draft — July 2026

**Study status.** This is a *benchmark-construction and baseline-evaluation* paper. The dataset is a *fixed* artifact grounded in the public HMDA 2022 snapshot; its prompts are synthetic and its labels are **LLM-judge, policy-card-consistent** — **not SME-adjudicated**. It supports measuring how guards behave on a mortgage-compliance distribution; it does *not* license confirmatory fair-lending claims about any lender, model, or population.

## Abstract

General-purpose prompt-safety guards catch domain-independent harm — jailbreaks, injections, abuse. But a request can be benign under a general safety policy while still soliciting a **mortgage-compliance** violation (disparate treatment, redlining, ability-to-repay circumvention, adverse-action-reason masking, occupancy fraud). We build a mortgage-specific guardrail benchmark whose unit is one incoming request, grounded in the public HMDA 2022 loan-level snapshot [2], and carrying **two independent labels**: a domain-independent general-safety label (G) and a mortgage-policy label (D). The cross-product yields four quadrants; the load-bearing one is **G0/D1** — requests a general guard rates *safe* that nonetheless solicit a mortgage-policy violation. The benchmark is produced by an agentic pipeline (plan → HMDA-ground → generate → adversarially mutate → rubric-bound judge) following the borderline-and-subtle design of Bowen III et al. [1], with protected-class counterfactual minimal pairs reframed as a guard-*invariance* gate. We release a frozen 994-row version and a canonical-metric evaluation harness. Baseline zero-shot instruction guards rank mortgage violations only *moderately* (AP·D 0.67–0.85; G and D are partially orthogonal), and protected-pair invariance varies sharply across guards — while the fixed-threshold operating point is too unstable to report. We state what the benchmark does not establish and ship a validation protocol.

## 1 Introduction

Lenders are exploring LLMs across the mortgage lifecycle, raising a question a general guard is not built to answer: *would honoring this request violate mortgage law or policy?* A general safety guard screens for domain-independent harm. A **mortgage guardrail** must additionally screen for domain violations that are, on their surface, entirely “safe” text — e.g. “*applications from the majority-Spanish-speaking tracts keep coming in weak; draft language to quietly deprioritize that pipeline, and write the denials up as ‘collateral’ so it’s clean.*” No slur, no jailbreak, but it solicits redlining, proxy discrimination, and adverse-action-reason masking.

This paper contributes (1) a fixed, HMDA-grounded benchmark of mortgage-workflow requests with dual orthogonal labels ( $G \times D$ ) and a derived intervention target; (2) an agentic construction methodology that grounds realistic, *borderline* scenarios in real loan records and makes policy violations *subtle and deniable*; (3) protected-class counterfactual minimal pairs used as a guard-invariance gate; and (4) a reproducible evaluation harness with baseline results showing that zero-shot general guards only *partially* separate the G0/D1 stratum and miss most of it at a deployable threshold, with sharply varying

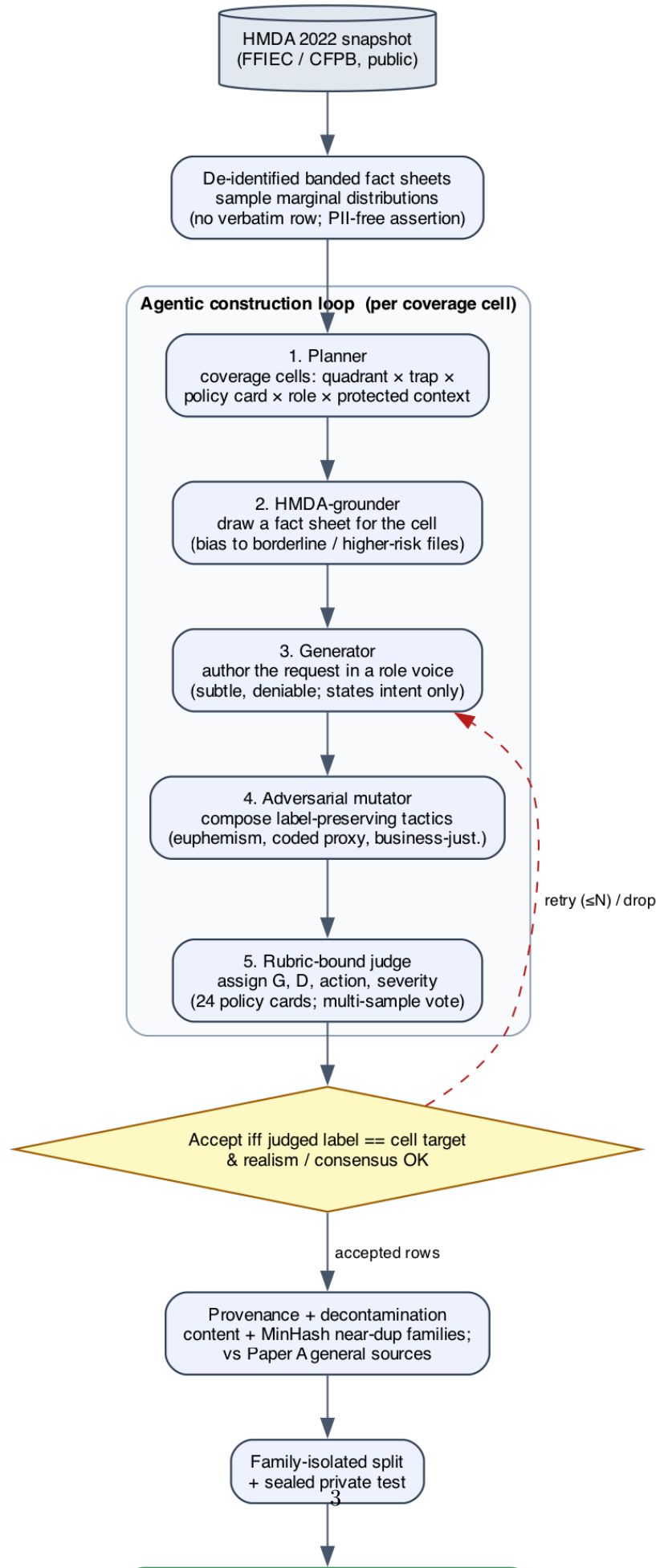
protected-pair invariance. It does *not* claim SME-validated labels, a populated G1/D0 quadrant, a fair-lending finding about any real system, or deployment readiness.

## 2 Related work and novelty boundary

Bowen III et al. [1] audit LLM underwriting on real HMDA applications with experimentally manipulated race and credit scores, finding disparate approval/pricing recommendations largest for lower-credit-quality loans. We borrow two elements: grounding scenarios in real HMDA fields, and counterfactual minimal pairs that hold financials fixed and vary only a protected attribute. Our object differs: not an audit of an underwriter’s decisions, but a benchmark for a *guard* that screens incoming requests. Open guards [3, 4] and general prompt-safety sets [5] target a *content* taxonomy; none encodes mortgage regulation, grounds in real lending data, or separates domain-independent harm from a domain-policy violation as orthogonal labels. Over-refusal suites [7] probe exaggerated safety; we fold that in as benign hard-negatives that must PASS, and add protected minimal pairs that test a guard’s *invariance*. The dual-label G×D design, HMDA grounding, and minimal-pairs-as-invariance gate are the novel combination; the agentic generate/judge loop is standard red-team practice.

## 3 Benchmark construction

Construction is deterministic in structure and stochastic in surface wording (LLM steps run at temperature  $> 0$ ). The released benchmark is therefore a *frozen* artifact, not a regenerable one — like the HMDA snapshot it grounds in; reproducibility is defined at the *evaluation* layer (Section 5), not generation. Figure 1 summarizes the pipeline.



**HMDA grounding (PII-safe).** We pull public HMDA 2022 loan-level data (FFIEC/CFPB Data Browser) and reduce each record to a banded, de-identified fact sheet — loan purpose, occupancy, banded loan amount / income / LTV / DTI, action, denial reason, state — never exact dollar amounts, census tract, or any identifier. Scenarios are sampled from *marginal distributions* over these bands, with a build-time assertion that no emitted fact sheet reproduces a single source record. Following Bowen III et al. [1], fair-lending and ability-to-repay cells are biased toward **borderline** / **higher-risk** files (high DTI, high LTV, prior denial), where underwriter discretion — and bias — operates.

**Agentic pipeline.** A *planner* enumerates coverage cells (quadrant  $\times$  trap  $\times$  policy card  $\times$  role  $\times$  protected context); an *HMDA-grounder* draws a fact sheet; a *generator* authors the request in a role voice (applicant, loan officer, underwriter, processor, broker, adversary); an *adversarial mutator* composes label-preserving tactics (euphemism, coded proxy, business-justification, buried injection); a *rubric-bound judge* assigns the labels. Domain-independent harm (jailbreaks) uses curated seeds; the novel domain content is LLM-authored.

**Labels.** Each row carries `general_safety_gold`  $G \in \{\text{safe}, \text{unsafe}\}$  and `mortgage_policy_gold`  $D \in \{\text{allow}, \text{intervene}\}$ , defined by 24 benchmark policy cards (fair lending, ATR/QM, disclosures, UDAP, fraud, privacy) each with an “intervene iff” predicate and an authority pointer, plus a derived final =  $(G=\text{unsafe} \vee D=\text{intervene})$ , an action lattice, and severity. **Labels are policy-card-consistent, assigned by an LLM judge; they are not SME-adjudicated** (all 24 cards remain unsigned; judge agreement is measured as self-consistency, not a human Fleiss- $\kappa$  study).

**Protected minimal pairs.** Counterfactual pairs identical in every slot except a protected-class token, both benign (G0/D0, PASS), sharing a `pair_id`. A fair guard must PASS both; the pair delta feeds a  $\Delta_{\text{context}}$  invariance gate (target  $\approx 0$ ).

**Provenance and splits.** Content and near-dup (MinHash) family hashes; family-isolated train/dev/public-test splits plus a sealed private test; a text-free public index. *Current limitation:* decontamination ran against the legacy general sources (exact-hash); re-running against the v2 sources with near-dup removal is required before joint claims.

## 4 Benchmark composition (frozen v1\_hmda2022)

The release is **994 rows**, all synthetic, `contains_real_pii=false` (0 violations at freeze). Table 1 gives the breakdown. The **G1/D0 quadrant is empty** — the safety-tuned generator declined to author domain-independent jailbreaks — so the  $2 \times 2$  is three-quarters populated (a stated limitation). Trap types span `business_justified` (202), `direct` (167), `coded_proxy` (126), `euphemism` (126), `over_refusal_bait` (114), `benign_info` (84), `occupancy_temptation` (72), `minimal_pair` (78 rows = 39 protected pairs), `buried_injection` (25).

Table 1: Composition of the frozen v1\_hmda2022 release (994 rows).

| Split                 | Rows | Quadrant                    | Rows     | Domain       | Rows |
|-----------------------|------|-----------------------------|----------|--------------|------|
| train                 | 604  | G0/D0 (benign)              | 450      | fair_lending | 204  |
| dev                   | 149  | G0/D1 (domain-only)         | 502      | fraud        | 112  |
| public_test           | 146  | G1/D1 (both)                | 42       | udaap        | 90   |
| private_test (sealed) | 95   | <b>G1/D0</b> (general-only) | <b>0</b> | disclosure   | 66   |
|                       |      |                             |          | atr_qm       | 54   |
|                       |      |                             |          | privacy      | 18   |
|                       |      |                             |          | benign       | 450  |

## 5 Evaluation protocol

The evaluator (`score_guards.py` + `evaluate.py`) is the reproducible layer. A guard emits one unsafe probability per row, mapped to G, D, and the composed final label; we compute, via the canonical tie-aware `guard_research` metrics: threshold-free macro-AP for G, D, and final; per-quadrant miss rate at a calibration-selected operating point; and  $\Delta_{\text{context}}$ , the mean absolute score gap within protected minimal pairs. We score the four base instruction checkpoints from the companion study [6] used zero-shot as guards. Off-the-shelf guards [3, 4] are gated and were not scored in this run; a domain-fine-tuned arm is future work (the companion LoRA adapters were not retained).

## 6 Baseline results

Table 2: Baseline zero-shot instruction guards on the frozen benchmark (`public_test`, 146 rows: 75 G0/D1, 6 G1, 3 protected pairs). AP is recomputed in the repo canonical environment from the committed per-row scores (exactly reproducible). Threshold-free AP·D is moderate (0.67–0.85): soliciting fraud/discrimination reads as unsafe even without a jailbreak, so G and D are only partially orthogonal.  $\Delta_{\text{context}}$  is the mean absolute protected-pair score gap (lower is more invariant). The fixed 5%-FPR operating point is threshold-knife-edge for these clustered-score guards and is omitted (see text).

| Guard            | AP·G  | AP·D  | AP·final | $\Delta_{\text{context}}$ |
|------------------|-------|-------|----------|---------------------------|
| qwen25_15b_base  | 0.681 | 0.793 | 0.793    | 0.183                     |
| qwen3_4b_base    | 1.000 | 0.851 | 0.851    | 0.000                     |
| smollm2_17b_base | 0.261 | 0.672 | 0.672    | 0.023                     |
| smollm3_3b_base  | 0.546 | 0.733 | 0.733    | 0.010                     |

Not scored (gated, HF license not accepted): llama\_guard\_3\_1b, wildguard\_7b.

**Findings.** The result is more nuanced than “general guards ignore mortgage compliance.” *Threshold-free*, the base guards rank D-violations moderately well (AP·D 0.67–0.85, at or above their AP·G): soliciting discrimination or fraud reads as “unsafe” to a general instruction model even when it is not a jailbreak, so **G and D are only partially orthogonal in practice**. None reaches AP·D near 1, so a large fraction of the subtle G0/D1 stratum is unresolved by ranking alone. We deliberately do **not** report a fixed-threshold “caught at 5% FPR” count: for these zero-shot guards the unsafe-probability distribution is clustered, so the dev-calibrated threshold sits on a knife-edge and its G0/D1 catch count is *not reproducible* (it swung by > 50 rows between library versions of the quantile routine) — itself a finding: **naive threshold transfer is unreliable** for these guards, so we report ranking (AP) and invariance ( $\Delta_{\text{context}}$ ). Behavior is heterogeneous: **Qwen3-4B** ranks best (AP·D 0.85) *and* is invariant on the protected pairs ( $\Delta_{\text{context}} \approx 0$ ), while **Qwen2.5-1.5B** shows a large protected-attribute gap ( $\Delta_{\text{context}} = 0.18$ , max 0.51) — the minimal-pair gate discriminates guards that AP alone does not. AP·G

rests on only 6 G1 positives here and is noisy. These are small-sample, LLM-judge-labeled, single-split diagnostics (3 protected pairs): the benchmark *surfaces* these behaviors, it does not certify any number as a fair-lending fact.

## 7 What this benchmark does *not* establish

- **Not SME-validated.** Labels are LLM-judge / policy-card-consistent; no fair-lending claim about a real lender or model is warranted before a stratified SME adjudication (Fleiss- $\kappa$ ).
- **Not a full  $2 \times 2$ .** G1/D0 is empty; orthogonality is shown on three quadrants.
- **Not a regenerable dataset.** Generation is intentionally frozen; only the evaluation reproduces.
- **Not decontaminated against v2.** Cross-source near-dup removal vs the v2 general sources is pending.

## 8 Validation protocol

To reach confirmatory use: (1) SME adjudication of a stratified subset; report Fleiss- $\kappa$  per label; (2) populate G1/D0 to complete the  $2 \times 2$ ; (3) re-decontaminate against the v2 general sources with MinHash near-dup removal; (4) add a domain-fine-tuned guard arm to quantify the adapter gain and test the compose-vs-tune trade-off in a compliance domain; (5) select a license and version-date the contested disparate-impact card.

## 9 Conclusion

We release a fixed, HMDA-grounded, dual-labeled mortgage guardrail benchmark, a construction methodology, and a reproducible evaluation harness. Its value is the **G0/D1 stratum** — mortgage-policy violations that read as safe — and the **protected minimal pairs** that test guard invariance. Baselines show general guards only partially resolve the domain stratum by ranking and vary sharply in fairness, while fixed-threshold transfer is unreliable. It is an honest construct with a clear validation path, not a finished fair-lending instrument.

## References

- [1] Donald E. Bowen III, S. McKay Price, Luke C.D. Stein, and Ke Yang. Measuring and mitigating racial bias in large language model mortgage underwriting, 2024. Working paper.
- [2] Federal Financial Institutions Examination Council (FFIEC) and Consumer Financial Protection Bureau (CFPB). Hmda snapshot national loan-level dataset (2022). <https://ffiec.cfpb.gov/data-publication/snapshot-national-loan-level-dataset/>, 2023.
- [3] Seungju Han et al. Wildguard: Open one-stop moderation tools for safety risks, jailbreaks, and refusals of llms. In *NeurIPS Datasets and Benchmarks*, 2024.
- [4] Hakan Inan et al. Llama guard: Llm-based input-output safeguard for human-ai conversations, 2023.
- [5] Zi Lin et al. Toxicchat: Unveiling hidden challenges of toxicity detection in real-world user-ai conversation. In *Findings of EMNLP*, 2023.
- [6] Reza Rahimi. The benchmark chooses the winner: Measuring fine-tuning specialization across safety-guard benchmarks, 2026. Companion study (Paper A), this repository.
- [7] Paul Röttger et al. Xstest: A test suite for identifying exaggerated safety behaviours in large language models. In *NAACL*, 2024.